

SUMMARY

Results-driven Computer Science undergraduate specializing in **full stack development, scalable backend engineering, and machine learning**. Proficient in designing **production-grade RESTful APIs, multi-agent LLM pipelines, and RAG-based systems** using **Python, Django, FastAPI, and React.js**. Hands-on experience with **cloud deployment (AWS, Render), containerization (Docker), CI/CD pipelines, and Agile SDLC**. Skilled at integrating **Generative AI, LLMs, and deep learning models** into end-to-end applications — from **model development and deployment to scalable system design**. Actively targeting full-time opportunities as a **Software Developer, Backend Developer, or ML / Data Science Engineer**.

EDUCATION

VIT Bhopal University

B.Tech in Computer Science & EngineeringCGPA: 9.13

Bhopal, Madhya Pradesh

Sep. 2023 – May 2027

SKILLS SUMMARY

- **Languages:** C++, Python, Java, SQL
- **Backend:** Django, Django REST Framework (DRF), FastAPI, Celery, Redis, RESTful API Design, JDBC, MVC, DAO Pattern, Multithreading
- **ML / AI:** Supervised & Unsupervised Learning, Deep Learning, NLP, LLMs, RAG, Prompt Engineering
- **Frameworks & Libraries:** Scikit-learn, PyTorch, TensorFlow, Pandas, NumPy, OpenCV, Sentence Transformers, Matplotlib, Streamlit
- **CS Fundamentals:** Data Structures & Algorithms, Operating Systems, DBMS, Object-Oriented Programming (OOP), Computer Networks
- **Databases:** MySQL, PostgreSQL (PostGIS), ChromaDB, SQLAlchemy, Alembic
- **Tools & Cloud:** Git, GitHub, Docker, AWS, Render, Vercel, CI/CD, Live Deployment, Jupyter Notebook, Google Colab, Tailwind CSS, Framer Motion

WORK EXPERIENCE

Artificial Intelligence Intern | Coding Jr.

Remote | Jun 2025 – Aug 2025

- Designed and optimized supervised **machine learning** models using **Python, Scikit-learn, and Pandas** for feature engineering, preprocessing, and hyperparameter tuning, achieving up to **15% accuracy improvement** over baseline benchmarks.
- Deployed and served trained models as scalable **RESTful APIs** on **Render and AWS**, reducing inference latency by **30%** and enabling production-grade, low-latency **model deployment** for end users.
- Engineered automated model evaluation pipelines using **Python, Scikit-learn, and Jupyter Notebooks**, cutting manual QA time by **40%** and accelerating research-to-deployment cycles within an **Agile** sprint workflow.
- Managed **version control** using **Git/GitHub**; collaborated cross-functionally to convert 3 research prototypes into production-ready deployments, improving team delivery velocity by **20%**.

PROJECTS

Prahari — Real-time Civic Intelligence & Response System

[GitHub] [Live] May 2025

Tech Stack: Python, Django, DRF, Celery, Redis, Django Channels, PostgreSQL (PostGIS), ChromaDB, Groq LLM API, Sentence Transformers

- Architected a real-time civic incident triage platform across **3 civic domains** (healthcare, legal rights, emergency incidents) with **scalable microservice architecture** and a **5-Agent LLM pipeline** (Groq LLaMA 3.3 70B / 3.1 8B) covering Sentinel, Triage, Legal Audit, Coordination, and Translation stages.
- Engineered a **RAG pipeline** using **ChromaDB and Sentence Transformers (all-MiniLM-L6-v2)** for semantic retrieval across **3 Indian legal frameworks** (BNS, CrPC, IPC) as dynamic LLM context; streamed live incident updates via **Django Channels (WebSockets / Daphne ASGI)**.
- Implemented **automatic LLM failover** (70B → 8B on HTTP 429), PostGIS/JSON GIS fallback, JWT authentication, multi-tenant isolation middleware, and a custom sliding-window **IP rate limiter** via **Celery and Redis**.

YuVA Wellness — AI-Driven Mental Health Platform

[GitHub] [Live] Jan 2026

Tech Stack: Python (FastAPI), React.js, PostgreSQL, Google Gemini API (1.5 & 2.0), Google OAuth 2.0, Tailwind CSS, Framer Motion

- Engineered a real-time **AI model-switching interface** toggling across **3 Gemini model variants** (1.5 Flash, 1.5 Pro, and 2.0) via **FastAPI** async endpoints, dynamically balancing response speed with deep emotional reasoning across 3 distinct user interaction modes.
- Implemented **Google OAuth 2.0 API integration**, multi-layer **bcrypt** password hashing, **JWT** session authentication, and a keyword-based **crisis detection** layer for proactive real-time distress intervention.
- Designed a wellness scoring algorithm over **PostgreSQL (SQLAlchemy + Alembic)** for real-time mood trend dashboards; custom **FastAPI** middleware maintained **99.9%** backend availability on Render.

Brain Tumor Segmentation — BraTS 2020

[GitHub] Apr 2025

Tech Stack: Python, PyTorch, 3D U-Net, Nibabel, NumPy, OpenCV, Matplotlib, Google Colab

- Implemented a **3D U-Net** deep learning architecture in **PyTorch** for multi-modal MRI brain tumor segmentation on the **BraTS 2020 benchmark (369 MRI scans)**, handling **4 modalities** (T1, T1ce, T2, FLAIR) with GPU acceleration on Google Colab.
- Applied intensity normalization, **3D patch-based augmentation**, and volumetric preprocessing using **Nibabel and NumPy** on high-dimensional MRI scans to improve model generalization and robustness.
- Evaluated segmentation quality using **Dice Coefficient and IoU** metrics across all tumor sub-regions; visualized 3D MRI slices and predicted segmentation masks using **Matplotlib and OpenCV**.

CERTIFICATIONS

- **Coursera — Computer Networking:** IP addressing, TCP/IP protocols, routing, DNS, network data transmission fundamentals, and OSI model concepts. [Certificate]
- **NPTEL — Machine Learning:** Core ML algorithms, supervised and unsupervised learning, model evaluation techniques, and hands-on Python implementation. [Certificate]
- **NPTEL — Marketing Analytics:** Data-driven marketing strategies, customer segmentation, predictive modeling, campaign performance analysis, and ROI optimization using statistical and machine learning techniques. [Certificate]